Unit 1 – Using Objects and Methods
Unit 1 Coding Project

Calculator and String Analyzer

You will be creating a Calculator and String Analyzer that performs a variety of mathematical and string-based operations using variables, arithmetic expressions, method calls, and object interactions. This project reinforces sequencing, basic objects, method calls, and math and string operations.

1. Create a `Calculator` Class: Define a `Calculator` class with the following:
   - Two instance variables: `double num1`, `double num2`
   - Two constructors: a default constructor and a parameter constructor
   - Methods:
     - `public double add()` – Returns the sum of `num1` and `num2`
     - `public double subtract()` – Returns the difference of `num1` and `num2`
     - `public double multiply()` – Returns the product of `num1` and `num2`
     - `public double divide()` – Returns the quotient of `num1` and `num2` (assume both are non-zero)
     - `public double power()` – Returns `num1` raised to `num2`
     - `public double absoluteDifference()` – Returns the absolute difference of `num1` and `num2`
     - `public double squareRootOfSum()` – Returns the square root of the sum of `num1` and `num2`
     - `public int randomBetweenNums()` – Returns a random number between `num1` and `num2` *(Hint: Use* `Math.min()` *or* `Math.max()` *to help)*

2. Create a `StringManipulator` Class: Define a `StringManipulator` class with the following:
   - Instance variables: `String text1`, `String text2`
   - Two constructors: a default constructor and a parameter constructor
   - Methods:
     - `public String combineStrings()` – Returns `text1 + text2`
     - `public int getCombinedLength()` – Returns the length of `text1` and `text2` added together
     - `public String firstHalf()` – Returns the first half of `text1`
       - If `text1` is JAVA, this method returns JA
       - If `text1` is HELLO, this method returns HE
     - `public int indexOfFirstLetter()` – Returns the index of the first letter of `text2` in `text1`
     - `public String swapFirstAndSecondHalf()` – If `text1` has an even length, swaps the two halves. If `text1` has an odd length, the middle letter stays with the second half of the word.
       - If `text1` is JAVA, this method returns VAJA
       - If `text1` is HELLO, this method returns LLOHE

3. Create the Main class to run the program
   - Ask the user for input (numbers for math operations and words for string operations).
   - Create objects of `Calculator` and `StringManipulator`.
   - Call methods from each object.
   - Display output with formatted results.

There are two sample runs of a working program below.

Sample Run #1	Sample Run #2
Enter in the first number: 10 Enter in the second number: 2 Enter first word: Computer Enter second word: APCSA Sum: 12.0 Difference: 8.0 Product: 20.0 Quotient: 5.0 Power: 100.0 Absolute Difference: 8.0 Square Root of Sum: 3.46410161… Random Number: 5 Combined: ComputerAPCSA Combined Length: 13 First Half: Comp Index of Letter: −1 Swapped: uterComp	Enter in the first number: 4 Enter in the second number: 6 Enter first word: Testing Enter second word: trust Sum: 10.0 Difference: -2.0 Product: 24.0 Quotient: 0.6666666… Power: 4096.0 Absolute Difference: 2.0 Square Root of Sum: 3.162277… Random Number: 4 Combined: Testingtrust Combined Length: 12 First Half: Tes Index of Letter: 3 Swapped: tingTes

4. After successfully coding your program, answer the following in complete sentences.

- For the method `squareRootOfSum()`, explain how you wrote your expression. What is another way you could write the method body and get the same result?
- For the method `randomBetweenNums()`, explain the algorithm you used to get a random number between the two values.
- For the method `swapFirstAndSecondHalf()`, explain why the algorithm you used works for either an even number of letters in text1 or an odd number of letters in text1.
- Describe a situation where calling a method returned an unexpected result. How did you fix it?

Submit the link to your program, as well as your answers to your 5 reflection questions, in the Google Classroom.